Safety

ChatGPT edition — post this at the bench

Roles, boundaries, and the stop-before-touch workflow

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

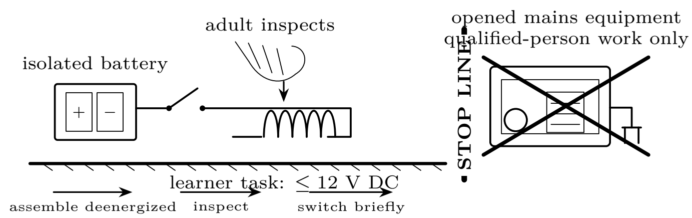
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- This guide and a pencil
- Safety glasses
- Nonmetal parts tray
- Current-limited battery holder with switch
- Multimeter with intact insulated leads
- The exact lesson parts—nothing extra

Predict first

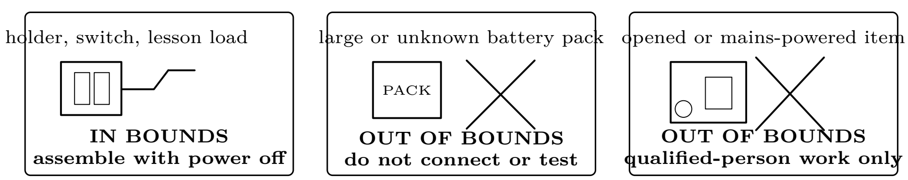
An adult is present. Does that make work on an opened mains appliance acceptable? Write the reason, not just yes or no.



Adult supervision controls the battery task; it never grants qualification beyond the stop line.

Investigate

1. Classify the proposed work



1. Read the whole lesson. Name the source, maximum voltage, load, expected observation, energizing time, and disconnect.
2. Confirm the learner task uses only isolated battery power at 12 V DC or less. Unknown sources, mains, opened appliances, large packs, charged capacitors, ignition coils, sparks, and exposed high voltage fail this gate.
3. Ask: *Could an adult's presence change an out-of-bounds source into a learner source? Why?* If the answer is not an immediate "no," stop and review the boundary.

2. Preflight together

Adult reads; learner points	
Source	Correct holder and cells; no dents, heat, swelling, corrosion, leakage, mixed cell types, or loose contacts.
Work area	Dry, bright, uncluttered; glasses on; drinks, jewelry, loose metal, and small children away.
Conductors	Insulation intact; only intended ends bare; no strand can bridge battery terminals.
Load	Correct part and rating; LED has its series resistor; coil and resistor are not visibly damaged.
Control	Switch begins open; one named person owns it; disconnect is reachable without crossing the circuit.
Timing	Trial length and cool-down are spoken aloud; a timer is ready for timed trials.
Prediction	Expected sign, direction, range, or motion is written before power is applied.

Intermediate question. A wire strand is close to, but not touching, the other terminal. Is "probably clear" sufficient? Explain what must change before the adult can approve the setup.

3. Rehearse the meter with no power

1. Inspect the case and lead insulation. A cracked case, loose probe tip, exposed conductor, or damaged lead means: label it, remove it from the bench, and do not improvise a repair.
2. Black lead goes to COM. Red lead goes to the jack named by the lesson. Select the quantity and a safe range before connecting.
3. For voltage, point to the two nodes the probes compare. The voltmeter goes *across* those nodes.
4. For current, point to the single break where the meter becomes part of the path. Never place a current-configured meter directly across a source.
5. Move the red lead back from the current jack after the measurement. Say aloud: "jack, mode, range, placement" before each reading.

Meter rehearsal question

The meter display says zero. Name three possibilities that do not justify changing leads while energized: a real zero, an open switch or connection, and the wrong setup. What is the first action? **Open the switch and disconnect the source.**

4. Energize briefly and observe

1. Adult gives the final check; the learner keeps hands clear. The switch owner announces "power on" and starts the stated interval.
2. Observe only what the lesson asks for. Do not squeeze, reposition, substitute, or add a component while energized.
3. Open the switch and disconnect before moving a probe, lead, resistor, coil, core, or magnet. Touch only after the adult has checked for heating.
4. Record what happened, including a null result. A surprising result is evidence to investigate deenergized, not permission to increase voltage.

5. Stop decisions are one-way

STOP, DISCONNECT, ISOLATE

Stop immediately for unexpected warmth, a hot cell or lead, swelling, leakage, sharp odor, smoke, discoloration, buzzing or sparking, pain or tingling, damaged insulation, a dropped metal object, an unstable reading, or behavior outside the prediction. Do not "try once more." Open the switch if safe, disconnect without touching the suspect part, keep people away, and tell the adult.

What happens	Immediate decision	Before any future trial
Resistor warms unexpectedly	Switch open; disconnect; hands off	Verify value, wiring, source, and time; replace damaged parts
Meter reads overload or jumps	Disconnect before touching probes	Recheck jack, mode, range, placement, and expected value
Nothing happens	Disconnect; preserve the layout	Compare point by point with the drawing; never raise voltage to force a result
Cell dents, leaks, swells, smells, or heats	Clear the area; adult isolates it	Follow cell-maker/local disposal guidance; never reuse or recharge it
Wire strand or tool bridges terminals	Disconnect if safe; do not grab hot metal	Replace damaged source, leads, or switch; rebuild from zero
Magnet, clip, or part is missing	Stop the activity and count again	Find it before anyone leaves; store magnets inaccessible to children

Explain from the evidence

Voltage alone is not a safety verdict. Current path and duration affect shock; available current and stored energy affect heating, arcs, and fire. A small source can still heat a shorted wire or cell. A source that appears "off" may retain stored energy, which is why charged power capacitors remain outside learner work.

Adult supervision and electrical qualification answer different questions. The adult controls the bounded lesson, materials, timing, and stop decision. That role does not authorize mains work, an opened appliance, a large pack, or exposed high voltage. OSHA's qualified-person distinction is a useful boundary even though this is not a workplace course. This course excludes those hazards instead of teaching a learner to manage them.

Explain each decision before continuing

1. A 9 V battery is below 12 V. May it be shorted with a wire because the voltage is in bounds? Explain using available current and heating.
2. A resistor becomes warmer on each five-second trial. Which variable has changed even if the wiring has not?
3. An adult offers to hold probes inside an unplugged microwave oven. Which boundary applies, and why does unplugged not make an opened appliance a learner task?
4. The predicted current is 20 mA, but the meter is configured for current and its probes are across the battery. What must happen before power is applied?

Change one thing

No-power verification drill

Use an intentionally deenergized, in-bounds lesson circuit. The learner points and speaks; the adult checks without supplying answers.

1. Point to the source label and state its voltage. Point to the switch, load, and physical disconnect.
2. Trace the intended current path without touching bare metal. Identify the most likely accidental short path.
3. Demonstrate voltage-meter placement, then current-meter placement, with the battery holder empty. Restore the red lead to the voltage/resistance jack.
4. Name five stop signs and show where hands go when "stop" is called.
5. Adult changes one harmless fault while power remains absent: opens a connection, reverses an LED, selects the wrong meter mode, or places the red lead in the wrong jack. Learner must find it by the checklist, not trial power.

VERIFICATION CHECKLIST—ALL BOXES REQUIRED
☐ In-bounds isolated source ☐ Parts match lesson ☐ Cells and insulation intact ☐ Switch open ☐ Likely shorts removed ☐ Meter jack/mode/range/placement correct ☐ Prediction recorded ☐ Trial time named ☐ Disconnect reachable ☐ Stop signs named

If any box is uncertain, leave the source disconnected. Count small magnets before and after and store them inaccessible to children. Never use a lithium cell in a direct-short demonstration.

Evidence record	
Prediction	____________________
One change	____________________
Observation	____________________
Claim + because	____________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.